Project 2

Macroeconomics 1

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Due: Wednesday, March 31, 2024, 00:00 P.M.

- Show your work and write your answers clearly. Use Genius Scan, CamScanner or similar apps if you plan on writing your homework using pen and paper. Using text editors (Word, LaTeX, R Mark-down, ...) is optional, but highly recommended.
- Name the file containing your answers as Project2LastName.pdf. They must be sent to ravanmh@gmail.com before the deadline.
- **It's highly recommended to start working on this problem set as soon as possible. Do not postpone it to a later time. You are going to have an assignment next week, which must be returned by the end of the week after it.**

McCall model with stochastic wages

Consider a version of the McCall model where agents' wages change over time on the job. We study a single, infinitely lived, risk-neutral household with discount rate β in partial equilibrium. The timing is as follows:

- Enter the period either as an unemployed worker or as employed worker with wage w.
- If unemployed, receives b and draws a wage offer with probability α.
- If employed, earns w and draws a new wage with probability λ.
- The wages are drawn from distribution $F(w) \in \Delta[0, B]$.
- Upon receiving an offer, they can choose to accept or reject it.
- If accepted, they will receive the wage next period.
- If rejected, would be unemployed next period.

Reservation wage is denoted by R. Answer the following questions:

- State the Bellman equation for an unemployed worker. Explain it in words, and show that

$$(1-\beta)U = b + \beta\alpha Q,$$

where U is the value of being unemployed, $V(w)$ is the value of being employed and receiving wage w, and

$$Q = \int_R^B (V(w') - U)\mathrm{d}F.$$

- State the Bellman equation of an employed worker. FOr the continuation value, keep in mind that the worker gets a new offer w' with probability λ, but can refuse $V(w')$ and instead choose U. Explain it in words, and show that

$$(1-\beta)V(R) = R + \beta\lambda Q.$$

- Find the reservation wage when $\lambda = \alpha$. Explain your finding.

- Show that $R > b$ when $\alpha > \lambda$. What is the intuition?

One period unemployment benefit

Consider the decision problem of the workers studied in previous section. Assume the worker can be in 3 states:

1. Employed, receiving wage w.
2. Unemployed after being employed last period, receiving b.
3. Unemployed for more than 1 period, receiving nothing.

While unemployed, the worker receives wage offers from distribution F. Jobs end with probability δ. Answer the following questions

- State the Bellman equations for each of the 3 possible states.

- Show that the reservation wage is independent of how long the worker has been unemployed.

McCall Search Model

Consider the simple McCall search model studied in class.

- **Blackwell conditions:** Verify that this problem satisfies the Blackwell sufficient conditions. Why is this an important step forward?

- **Properties of the value function:** Specify the properties of value function, i.e. is it continuous, monotone, quasi-concave and differentiable?

- **Triangular wage offers:** Assume that the wage offers have triangular distribution with mean μ and $w \in (\mu - r, \mu + r)$. Find the variance of this distribution, and write down a code two simulate two such distributions for $\mu = 7$ and $r = 3, 5$. Calculate variance of each wage offer.

- **Value function iteration:** Write a code to find the value function as well as the optimal policy function using value function iteration method. Try each of the following functions as initial guess, and report the number of iterations needed for convergence.

$$v^0 = 1,\ w,\ w^2,\ \log w,\ \exp(w),\ \sinh(w)$$

- **Comparative statics:** Provide analytical solution for each of the following questions. Then, provide numerical evidence to support your arguments by plotting reservation wage as a function of the variable of interest.
  - Discount rate (β)
  - Social welfare program (c)
  - Increase in average offers (μ)
  - Mean preserving increase in risk (r)